

Requirements and Architectural Design for a Dual-Curriculum (GES/NaCCA and ABEKA) School Management System in Ghanaian Private Basic Education

Architectural Design Paradigm and Database Schemas

Private pre-tertiary education in Ghana operates within a dynamic, highly regulated landscape. To remain competitive and serve diverse student populations, many premium private basic schools (covering creche, nursery, kindergarten, primary, and junior high school) adopt a dual-curriculum model.¹ These institutions run the local Ghanaian curriculum, developed by the National Council for Curriculum and Assessment (NaCCA) and managed by the Ghana Education Service (GES), alongside the American-based, faith-focused ABEKA curriculum.¹ Integrating these distinct paradigms into a unified School Management System (SMS) presents unique software engineering challenges. The platforms must reconcile divergent philosophies of learning progression, cognitive evaluation models, grading structures, and operational cycles.³

A robust system architecture for a dual-curriculum SMS must support multi-curriculum partitioning at the database layer. This partitioning ensures that data schemas can capture standards-based progress metrics for the Ghanaian curriculum while simultaneously tracking conventional percentage-to-GPA records for ABEKA.³ By utilizing a polymorphic database schema, a single student identity can link to parallel curriculum tracks.⁷



The database design must accommodate specific regulatory and curriculum identifiers, such as national identity indicators (Ghana Card ID) ¹⁰, international student IDs ⁷, and structured classroom mappings.¹⁰ The following schema illustrates how a relational database can represent this dual-track structure:

SQL

```
-- Core student registry supporting national and curriculum-specific identifiers
CREATE TABLE students
(
    student_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    first_name VARCHAR 100 NOT NULL,
    last_name VARCHAR 100 NOT NULL,
    date_of_birth DATE NOT NULL,
    gender VARCHAR 10 CHECK (gender IN 'Male' 'Female'),
    ghana_card_id VARCHAR 20 UNIQUE -- National identification for statutory compliance
    nationality VARCHAR 50 DEFAULT 'Ghanaian'
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
)

-- Definition table for different curriculum tracks supported by the institution
CREATE TABLE curriculum_tracks
(
    track_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    track_name VARCHAR 50 UNIQUE CHECK (track_name IN 'GES_NACCA' 'ABEKA')
    is_active BOOLEAN DEFAULT TRUE
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
)

-- Dynamic enrollment association table linking students to curriculum-specific classrooms
CREATE TABLE student_enrollments
(
    enrollment_id UUID PRIMARY KEY DEFAULT gen_random_uuid()
    student_id UUID REFERENCES students(student_id) ON DELETE CASCADE
    track_id UUID REFERENCES curriculum_tracks(track_id) ON DELETE RESTRICT
    grade_level VARCHAR 50 NOT NULL -- Mapped to 'Basic 1-9' or 'Grade 1-9'
    academic_year VARCHAR 20 NOT NULL -- Format: '2025/2026'
    enrollment_date DATE NOT NULL DEFAULT CURRENT_DATE
    CONSTRAINT unique_student_session_track UNIQUE (student_id, academic_year, track_id)
)
```

This normalized database structure allows school administrators to query student rosters by grade level or curriculum track.³ Consequently, teachers can manage independent gradebooks and student portfolios without data collision.³

Unified Academic Calendars and Registries (Creche to Junior High School)

To manage dual schedules within a single institution, the academic calendar module must align two distinct operational cycles: the termly structure mandated by GES/NaCCA³ and the quarterly, lesson-by-day framework of the ABEKA model.⁶

The GES/NaCCA calendar is traditionally built on a three-term cycle (Term 1, Term 2, and Term 3), spanning from September through July or August.³ This schedule is aligned with the Education Strategic Plan (ESP 2018–2030) and is subject to national scheduling directives.³ In contrast, ABEKA relies on a standardized 180-day instructional year divided into four quarters (9-week periods).² ABEKA coordinates daily lessons using highly structured, predefined guides that require teachers to complete specific content and oral readings on specific days.⁶



To coordinate these parallel tracks, the system's scheduling engine must support nested calendar views. This design ensures that term-based examinations and national assessments, such as the National Standard Assessment Test (NSAT) at Basic 8 (JHS 2), can be scheduled alongside ABEKA's portfolio submission milestones.³

This complexity is also reflected in the student registry, which spans early childhood education to Junior High School.⁴ The system must maintain consistent academic records across these

developmental phases.³

Educational Phase	Ghanaian Grade Designation	ABEKA/US Grade Level	Core Pedagogical Focus & Media Requirements
Early Childhood Education (ECE)	Creche & Nursery ¹³	Preschool (K3 - K4) ⁶	Play-based milestone tracking; developmental psychology mapping. ¹³
Kindergarten	Kindergarten 1 & 2 (KG1 - KG2) ⁴	Kindergarten (K5) ²	Foundational literacy, numeracy, play-based learning, and early cursive writing. ²
Lower Primary	Basic 1 to Basic 3 (B1 - B3) ⁹	Grade 1 to Grade 3 ²	Standards-Based Curriculum (4Rs: Reading, wRiting, aRithmetic, cReativity). ⁹
Upper Primary	Basic 4 to Basic 6 (B4 - B6) ⁹	Grade 4 to Grade 6 ⁶	Continuous evaluation, learner-centered pedagogy, standalone History and French. ⁴
Junior High School	Basic 7 to Basic 9 (B7 - B9)	Grade 7 to Grade 9 ⁶	Common Core Programme (CCP) ³ , early high-school GPA transcript generation. ³

For early childhood stages, the system must track physical and psychological development metrics (such as nutrition, security, emotional maturity, and motor skills) rather than quantitative numerical grades.¹³ For the kindergarten and early primary levels (KG1 to Basic 2), the registry must support cursive handwriting monitoring and oral reading verification, which are mandatory components of the ABEKA program.²

As students progress to the Upper Primary and JHS phases, the system must handle more complex academic profiles, including standalone subjects like History, French, and Religious and Moral Education (RME).³

Dual-Track Assessment and Grading Engines

Designing a grading engine for a dual-curriculum school requires reconciling two fundamentally different approaches to educational evaluation: the competency-oriented, qualitative standards-based approach of NaCCA and the quantitative, point-accumulation GPA model of ABEKA.³

The GES/NaCCA SBC and CCP Grading Module

Ghana's basic education assessment model shifted from an objectives-based curriculum to a Standards-Based Curriculum (SBC) in 2019, followed by the Junior High School Common Core Programme (CCP) in 2020.¹

The primary school curriculum (Basic 1 to Basic 6) evaluates students based on core competencies.⁴ These include critical thinking, problem-solving, creativity, digital literacy, and cultural identity.

The JHS phase (Basic 7 to Basic 9) relies on a structured, multi-dimensional assessment model comprising School-Based Assessment (SBA) and end-of-term examinations.³ Under this framework, learning behaviors are assessed using specific profile dimension weightings. In basic mathematics, for example, the learning profile is distributed across three key areas¹⁹:

- Knowledge and Understanding: 30%¹⁹
- Application of Knowledge: 40%¹⁹
- Attitudes, Values, and Process Skills: 30%¹⁹

The final grade is calculated as a weighted combination of the continuous SBA score (which includes projects, portfolios, and community service tasks) and the terminal examination score, typically using a 60 : 40 or 50 : 50 ratio.³

At the system level, learning indicators must be logged using action-oriented performance levels (e.g., "Exceeding," "Meeting," "Approaching," or "Developing" standards).⁴ This design ensures that quantitative marks are mapped directly to qualitative competency descriptions.



83 –	B	3.00	3.00	Above Average ²
80 –	B-	2.75	3.00	Above Average ²
77 –	C+	2.50	2.00	Average ²
73 –	C	2.25	2.00	Average ²
70 –	C-	2.00	2.00	Average ²
67 –	D+	1.75	1.00	Below Average ²
63 –	D	1.50	1.00	Below Average ²
60 –	D-	1.25	1.00	Below Average ²
0 –	F	0.00	0.00	Failing ²

The system uses these mappings to calculate the student's cumulative grade point average.⁷
For standard classes, unweighted GPA is calculated as:

$$\text{GPA}_{\text{unweighted}} = \frac{\sum_{i=1}^n (\text{Grade Points}_i \times \text{Credits}_i)}{\sum_{i=1}^n \text{Credits}_i}$$

7

For advanced prep tracks, the engine can be configured to calculate weighted GPA, adding 0.5 grade points for honors classes and 1.0 point for advanced courses⁷:

$$\text{GPA}_{\text{weighted}} = \frac{\sum_{i=1}^n ((\text{Grade Points}_i + w_i) \times \text{Credits}_i)}{\sum_{i=1}^n \text{Credits}_i}$$

7

The system must maintain separate gradebooks for each curriculum track to support these distinct grading requirements.³ This structure allows the registrar to compile terminal reports for the GES district directorate and official transcripts for ABEKA Academy validation audits.³

Billing, Fee Management, and Local Payment Gateway Integrations

Managing school fees, canteen bills, and material charges in private schools requires a highly reliable financial module.¹¹ In Ghana, mobile money is the primary channel for digital financial transactions, making local payment gateway integrations essential.¹¹

To streamline this process, the billing module should integrate with prominent local payment aggregators.¹¹ These third-party gateways simplify integration by consolidating all major mobile money networks—MTN Mobile Money, Telecel Cash (formerly Vodafone Cash), and AT Money (formerly AirtelTigo Money)—into a single API endpoint.¹¹

Payment Gateway	Supported Mobile Money Networks	Setup & Maintenance Fees	MoMo Transaction Fee	Bank Settlement Period	Key Regulatory Compliance Standard
Paystack	MTN MoMo, Telecel Cash, AT Money ¹¹	Free ¹¹	1.95% ¹¹	T+1 Days ¹¹	Bank of Ghana Payment Service Provider, PCI-DSS Level 1 ¹¹
Hubtel	MTN MoMo, Telecel Cash, AT Money ¹¹	Free ¹¹	1.95% (volume discounts available) ¹¹	T+1 Days or Instant ¹¹	Bank of Ghana Payment Service Provider, ISO 27001 ¹¹
Flutterwave	MTN MoMo, Telecel Cash, AT Money, international cards ¹¹	Free ¹¹	Local: 1.95% — ¹¹	T+1 Days ¹¹	Bank of Ghana Payment Service Provider, PCI-DSS Level 1 ¹¹
expressPay	MTN MoMo,	One-time setup fee	~	T+2 Days (Instant)	Bank of Ghana

	Telecel Cash, AT Money, local cards ¹¹	possible ¹¹	(capped transfers) ¹¹	options) ¹¹	Payment Service Provider ¹¹
--	---	------------------------	-------------------------------------	------------------------	--

Payment Aggregator KYC and Onboarding

Before a private school can transition from sandbox to live transaction environments, it must provide verified compliance documents to meet the Anti-Money Laundering (AML) and Know Your Customer (KYC) requirements of the Bank of Ghana ¹¹:

1. **Corporate Business Registration:** Certificate of Incorporation and Certificate to Commence Business, issued by the Registrar General's Department. ¹¹
2. **Ghana Revenue Authority Taxpayer Identification Number (GRA TIN).** ¹¹
3. **Active Corporate Bank Account Details:** Verified checking account registered under the school's legal name for receiving automated bank settlements. ¹¹
4. **Proprietor and Director Identification:** Valid national identification (Ghana Card) for all registered business owners and academic directors. ¹¹
5. **Verified Physical Address:** Certified physical location details, often supported by a Ghana Post GPS digital address code. ¹¹

Cryptographic Security and Callback Mechanisms

To prevent floating-point rounding issues when calculating transaction balances, all monetary figures processed by the system must be registered and handled in **pesewas** (the smallest

currency division in Ghana, where $\text{GHS } 1.00 = 100 \text{ pesewas}$).²³

To initiate a billing transaction (e.g., school fees payment), the system's backend issues a secure POST request containing payment metadata to the gateway provider ²³:

JSON

```
// POST Request directed to the Payment Initialization Endpoint (e.g.,  
https://api.paystack.co/transaction/initialize)
```

```
{
```

```
  "email" "guardian.parent@domain.com"
```

```
  "amount" 250000 // Represents GHS 2,500.00
```

```
  "currency" "GHS"
```

```
  "reference" "SCH-FEE-2026-TERM1-98402"
```

```

"callback_url" "https://portal.school.edu.gh/api/v1/payments/callback"
"metadata"
"student_id" "9a3821ef-d2c3-48b1-bf7e-e2890a1b2c3d"
"invoice_id" "INV-2026-T1-0145"
"billing_item" "First Term School Fees"

```

The gateway returns a secure payment page URL. After the parent authorizes the payment on their mobile device by entering their mobile money PIN, the gateway dispatches an asynchronous payment notification (webhook) to the school's server.²³

To protect the webhook endpoint from spoofing attacks, the system must verify the cryptographic signature sent in the request header.²³ This verification is performed by generating a local Hash-based Message Authentication Code (HMAC) signature using the SHA-512 algorithm and comparing it to the received signature²³:

$$\text{Signature}_{\text{computed}} = \text{HMAC-SHA512}(\text{Request Body JSON Payload}, \text{Gateway Secret API Key})$$

23

JavaScript

```

// Node.js Express Secure Webhook Handler demonstrating HMAC Signature Verification
const crypto = require('crypto')

```

```

app.post('/api/v1/webhooks/payment-callback' (req, res) =>
  // Generate computed HMAC signature from request body
  const computedSignature = crypto.createHmac('sha512'
    process.env.PAYMENT_GATEWAY_SECRET_KEY
    .update(JSON.stringify(req.body))
    .digest('hex')

```

```

  // Compare computed signature with the signature in the request header
  if (computedSignature !== req.header('x-payment-provider-signature')
    return res.status(401).send('Cryptographic signature verification failed.'

```

```

const transactionEvent = req.body

```

```

    if (transactionEvent.event === 'charge.success')
      const transactionRef = transactionEvent.data.reference
      const totalAmountPesewas = transactionEvent.data.amount
      const studentId = transactionEvent.data.metadata.student_id
      const invoiceId = transactionEvent.data.metadata.invoice_id

      // Perform transactional database update
      db.transaction(async => {
        // Enforce idempotency: Verify if this transaction reference has already been processed
        const paymentRecord = await findOneOrNone('SELECT id FROM transactions WHERE
reference = $1' )

        if (!paymentRecord) {
          // Record the unique financial ledger transaction
          await findOne('INSERT INTO transactions (reference, student_id, invoice_id,
amount_pesewas, status) VALUES ($1, $2, $3, $4, $5)'
)
        }

        // Deduct balance from invoice in a single operation
        await findOne('UPDATE invoices SET outstanding_balance_pesewas =
outstanding_balance_pesewas - $1 WHERE invoice_id = $2'
, [totalAmountPesewas, invoiceId])
      })

      // Return HTTP 200 OK to acknowledge receipt of the webhook event
      res.sendStatus(200)

```

Webhooks can sometimes be delivered out of order or multiple times by payment aggregators. The webhook handler must enforce strict **idempotency** by checking for the unique reference string in the database before applying any financial credits.²³

Furthermore, because redirect URL parameters on the client browser can easily be manipulated by malicious users, the system must never update a payment status based solely on client-side redirects.²³ The system must always verify the status of a transaction directly by calling the aggregator's verification endpoint (e.g., GET /transaction/verify/:reference) on the server backend before rendering a successful receipt.²³

Developers can also integrate specific bank query operations. For instance, the system can fetch a list of supported banks by sending a GET request with the filter pay_with_bank=true to dynamically populate bank transfer options.²² For temporary transaction links, the system can configure the account_expires_at parameter in ISO 8601 format, ensuring transaction windows expire appropriately within a 15-minute to 8-hour range.²²

Administrative Reporting Module for NaSIA and GES Licensing

Private pre-tertiary schools in Ghana operate under the oversight of the National Schools Inspectorate Authority (NaSIA), established under the Education Regulatory Bodies Act, 2020 (Act 1023).²⁴ Section 98 of Act 1023 mandates that all pre-tertiary education institutions register and license their operations.²⁵ Failure to obtain an active operational license results in administrative sanctions, fines, or closure of the facility.²⁶



NaSIA inspectors evaluate schools using a standardized, four-point assessment scale.²⁷ This metric is divided into qualitative and operational performance ratings ²⁷:

NaSIA Scoring Tier	Performance Metric (Part A Evaluation)	Operations Compliance (Part B Evaluation)	Licensing Implications
Grade 4	Outstanding ²⁷	Exceeds Minimum Requirements ²⁷	Full operating license issued, low inspection frequency. ²⁸

Grade 3	Good ²⁷	Fully Meets Requirements ²⁷	License active; scheduled routine inspection updates. ²⁸
Grade 2	Satisfactory ²⁷	Partially Meets Requirements ²⁷	Provisional license issued with corrective action recommendations. ²⁸
Grade 1	Unsatisfactory ²⁷	Fails to Meet Requirements ²⁷	License denied, operational sanctions, possible closure. ²⁶

To help schools prepare for and maintain these licensing standards, the SMS must include an automated compliance exporter.²⁶ This module compiles data across several operational requirements outlined in the NaSIA inspection guidelines²⁶:

1. New School Establishment Tracking (Part A and Part B Compliance)

The system must log the school's progression through NaSIA's multi-step licensing workflow²⁸:

- **Part A (Authorisation to Establish):** Tracks the submission of the Expression of Interest (Eoi) form, the acquisition of the school establishment manual, and the verification of business registration documents from the Registrar General's Department.²⁴
- **Part B (Authorisation to Operate):** Records the submission of the Notice of Intent to Operate (Noi) form, scheduling for physical compliance inspections, and the receipt of provisional licensing logs.²⁸

2. Teacher Professional Licensing Logs

The system must maintain professional profiles for all teaching staff, including their registration status with the National Teaching Council (NTC).²⁶ The compliance module tracks each teacher's professional development status—categorizing staff as licensed teachers, induction teachers, or unlicensed temporary staff—and flags upcoming credential renewal dates.²⁶

3. Student-Teacher and Classroom Ratios

The platform must generate structural statistics that show pupil-to-teacher and pupil-to-classroom ratios across all instructional levels (KG, Primary, and JHS).⁴ These reports help schools demonstrate compliance with GES staffing and facility capacity standards.²⁶

4. Infrastructural Resource Registries

The database must maintain records of physical assets, such as science laboratories, computer labs, library holdings, first-aid resources, and sports areas.¹⁶ The compliance module links these records to the school's structural inventory, helping administrators verify that facilities meet baseline standards prior to physical inspections.²⁴

5. Police Clearances and Health Screenings

To comply with NaSIA licensing criteria, the system must log police clearances and valid medical certificates for all employees, especially those in child-facing roles.²⁶ The compliance engine flags any records that require updating.²⁶

Statutory Compliance Framework under the Ghana Data Protection Act, 2012 (Act 843)

A dual-curriculum school management system handles a large volume of sensitive personal data, including children's academic performance, behavioral records, health information, parental contact details, and financial histories.¹⁰ In Ghana, this processing is strictly regulated under the **Data Protection Act, 2012 (Act 843)**.²⁹

Under Act 843, the school serves as the **Data Controller** (defining the purpose and manner of processing), while the platform developer operates as the **Data Processor** (processing data on behalf of the controller).¹⁰ Both entities must register with the Data Protection Commission (DPC).²⁹



Data Security and Cryptographic Standards

The system must implement appropriate technical and organizational measures to prevent unauthorized access or processing of personal data.²⁹ This includes encrypting data in transit using TLS 1.3 and encrypting data at rest using AES-256.¹⁰ These measures protect all student files, payment registers, and system logs.¹⁰

Parental Consent and Special Personal Data

Under Act 843, any data relating to children is classified as **Special Personal Data**.²⁹ Processing special personal data is prohibited unless the school obtains explicit, verifiable consent from a parent or legal guardian.²⁹

For children under 13 years of age, the SMS must enforce a parental consent flow before collecting or processing any personal information¹⁰:

1. **Verifiable Consent Workflows:** During student enrollment, the system must send an OTP or signature-request link to the verified mobile number or email of the parent or guardian.²⁹ The platform logs the time, date, IP address, and parent identification credentials to provide a clear audit trail for the DPC.
2. **Granular Opt-In Controls:** Parents must be provided with granular opt-in checkboxes, separating educational processing (e.g., grading, attendance) from third-party sharing or promotional activities.¹⁰

Data Subject Rights

The software must provide parent-facing portals with self-service tools that allow parents to exercise their legal rights under Sections 18–35 of Act 843¹⁰:

- **Right of Access:** Parents can export a structured, machine-readable PDF or JSON file of all personal data held on their child (Data Portability).¹⁰
- **Right to Rectification:** Self-service correction workflows allow parents to request or directly execute updates to outdated contact numbers, medical alerts, or address profiles.¹⁰
- **Right to Erasure (Right to Be Forgotten):** When a student transfers to another institution, parents can initiate an erasure request. However, the system must balance this right with statutory academic archiving rules dictated by the GES.¹⁰

Cross-Border Cloud Hosting and Localisation

Under Act 843, personal data of Ghanaian citizens cannot be transferred to countries or stored on cloud servers outside Ghana unless the destination country has "adequate" data protection laws that meet or exceed Act 843 standards, or DPC-approved Standard Contractual Clauses (SCCs) are in place.²⁹

If the SMS is hosted on international cloud infrastructure (e.g., AWS, Azure, or Google Cloud), the system architecture must implement several measures²⁹:

1. **DPC-Approved Contractual Clauses:** Data processing agreements must include

specific clauses approved by the DPC.²⁹

2. **Transfer Impact Assessments (TIAs):** The platform operator must conduct TIAs before onboarding foreign cloud hosting partners.²⁹
3. **Local Hybrid Storage Options:** The database engine should support hybrid configurations, allowing schools to store sensitive biometric or personal identifiers on localized servers within Ghana, while using public cloud environments for non-sensitive transaction routing.

Data Retention and Secure Deletion Engine

To remain compliant, the SMS cannot store personal data indefinitely.²⁹ The system must run a cron-based data retention engine that automatically flags, archives, or securely deletes data categories based on standardized schedules¹⁰:

Data Classification Category	Statutory Retention Period	Processing Action Upon Expiry	Primary Regulatory Mandate
Student Academic Records	Permanent	Maintained in locked, secure archive table	GES Archiving Policy ¹⁰
Student Attendance Records	7 Years post-exit	Anonymized database scrub, deletion of logs	Institutional Audits ¹⁰
Financial / Payment Records	7 Years	Complete deletion from active table, archived for tax	Ghana Revenue Authority Audit Rules ¹⁰
User Account & Technical Logs	Duration of Account + 2 Years	Secure physical-level deletion	DPC Security Directives ¹⁰
Support Communications	3 Years from resolution	Purged from system databases	Act 843 Minimisation Principle ¹⁰
Marketing Consent Records	Duration of Consent + 3 Years	Deletion from notification rosters	Section 20 Consent Expiration ¹⁰

Mandatory Breach Notification Protocol

Under Section 31 of Act 843, if a data breach occurs and there are reasonable grounds to believe student or guardian data has been accessed by an unauthorized party, the school must notify both the Data Protection Commission and the affected individuals.²⁹

The SMS backend must incorporate an automated Security Incident Response workflow. Once an unauthorized data access event is confirmed, the system must trigger a countdown:

Breach Event Confirmed → Auto-Assemble Audit Payload → Dispatch DPC Notification ($\leq$ 72 Hours)

10

The incident response script compiles a detailed payload—including the estimated volume of affected records, the nature of the breach, immediate mitigation measures taken, and contact details for the school's certified Data Protection Supervisor—and dispatches it directly to the DPC.²⁹

Automated Parent Communication and Notification Routing

Effective parent communication is essential for managing academic activities and billing updates.²³ To accommodate varying access to digital connectivity among parents, the system must support an omnichannel communication engine.²⁹ This engine must dynamically route messages across WhatsApp, traditional Bulk SMS, and USSD channels to ensure inclusive reach.²⁹



WhatsApp Business API Integration

The SMS uses the WhatsApp Business API for rich-media communications and structural document distribution. This channel is ideal for ⁶:

- **Interactive Portfolios:** Sending PDF copies of terminal reports, ABEKA progress records, and NTC-approved schemes of learning.⁶
- **Template-Driven Consent:** Distributing structured permission requests (e.g., for school excursions) with single-tap interactive buttons. This allows parents to provide verifiable consent that is logged directly by the system's database.¹⁰

Bulk SMS Gateway and Emergency Fallback

For parents without active internet connectivity, the communication engine must automatically fall back to traditional SMS.²⁹ The SMS gateway is optimized for real-time notifications, including ²⁹:

- **Real-time Attendance Alerts:** Notifying parents immediately when a student is marked absent during morning roll call.
- **Transactional Payment Logs:** Sending instant SMS confirmations after school fee payments are verified via mobile money webhooks.²³
- **Emergency Notifications:** Dispatching urgent alerts (e.g., severe weather closures) to all parents simultaneously.

Offline-Capable USSD Portal

To support parents who use basic mobile phones or operate in areas with poor data coverage, the system must include a lightweight USSD (Unstructured Supplementary Service Data) application.²⁹ Parents can dial a dedicated short code (e.g., *843*99#) to check student balances, view grades, and update notification preferences over the standard GSM cellular network:

Parent Dialed Code: *843*99#
Academic Progress Summary
Check Outstanding Invoice
Confirm Field-Trip Consent
Dynamic Alert Preferences
Selection:

Output: "Ama Osei (ID: 28402) has an outstanding balance of GH¢1,200.00."

To authorize payment via MTN MoMo, dial *170# or pay in the portal.

By querying the database in real time, the USSD controller ensures that critical academic and financial metrics remain accessible to every parent, regardless of device type or geographic location.²⁹

Conclusions and Implementation Recommendations

To deploy a dual-curriculum school management system successfully within Ghanaian private basic schools, engineering and administrative teams should consider the following technical guidelines:

1. **Polymorphic Database Core:** Implement a polymorphic database architecture that isolates student identity details from curriculum-specific courses, grades, and records. This design prevents data conflicts between standard-based GES/NaCCA evaluations and points-based ABEKA GPA metrics.³
2. **Idempotent Payment Workhooks:** Ensure the payment module converts all transaction amounts into pesewas prior to processing.²³ Webhook handlers must use HMAC-SHA512 verification and implement strict database idempotency checks to prevent duplicate transactions.²³
3. **NaSIA Readiness Integration:** Build an automated reporting engine that maps school databases directly to NaSIA's inspection rubrics.²⁶ This ensures real-time compilation of teacher credentialing, pupil-teacher ratios, and facility utilization logs, helping schools maintain compliance and protect their license to operate.²⁴
4. **Privacy by Design (Act 843):** Incorporate strict compliance workflows under Act 843, including verifiable parental consent mechanisms for students under 13¹⁰, DPC-certified supervisor roles³⁰, local-hybrid cloud hosting, and automated data retention policies.¹⁰
5. **Offline Communication Access:** Deploy a hybrid communication gateway that uses WhatsApp for rich media, Bulk SMS for real-time transactional alerts, and USSD queries for offline parental access.²³ This multi-channel approach ensures that all parents can stay informed and engage with the school, regardless of their device type or data connectivity.²⁹

Works cited

1. Ghana's Standards-Based Curriculum Reform: National Insights from the Bono East Region Experience - ERIC, accessed June 3, 2026, <https://files.eric.ed.gov/fulltext/EJ1494172.pdf>
2. Academics - Indiana Christian Academy, accessed June 3, 2026, <https://indianachristianacademy.org/academics>
3. NaCCA JHS & SHS Curriculum Overview | PDF - Scribd, accessed June 3, 2026, <https://www.scribd.com/document/691860630/Understanding-the-New-GES-JHS-and-SHS-Syllabus-NaCCA-Curriculum-and-the-Common-Core-Programme-C>

[CP-2023-Online-Portal](#)

4. GHANA: CURRICULUM, PEDAGOGY, AND ASSESSMENT AT THE SERVICE OF THE RIGHT TO EDUCATION - Amnesty International, accessed June 3, 2026, <https://www.amnesty.org/en/wp-content/uploads/2026/01/IOR4006422026ENGLISH.pdf>
5. Implementation of Ghana's Standards-Based Curriculum by Religious and Moral Education Teachers in the Kwahu West Municipality - Noyam Journals, accessed June 3, 2026, https://noyam.org/?sdm_process_download=1&download_id=12856
6. Academics FAQs - Abeka | Dashboard, accessed June 3, 2026, <https://academy.abeka.com/Help/AcademicsFAQs.aspx>
7. Creating Homeschool Transcripts - Abeka, accessed June 3, 2026, <https://www.abeka.com/blog/creating-homeschool-transcripts/>
8. Info for Transcript or Report Card - Cheerful Heart Academy, accessed June 3, 2026, <https://www.cheerfulheartacademy.com/grades>
9. Ghana Primary School Curriculum Overview | PDF - Scribd, accessed June 3, 2026, <https://www.scribd.com/document/904958526/Ghana-Primary-Basic1-6-Curriculum-Summary>
10. Privacy Policy | EDVES Global - AI-Powered Pedagogy Platform, accessed June 3, 2026, <https://edves.com/privacy-policy.php>
11. MTN MoMo Website Integration Ghana: Costs & How-To, accessed June 3, 2026, <https://websysgh.com/how-website-development-companies-in-ghana-integrate-mobile-money-payment-systems/>
12. mathematics common core programme (ccp) - NATIONAL COUNCIL FOR CURRICULUM & ASSESSMENT, accessed June 3, 2026, <https://nacca.gov.gh/wp-content/uploads/2023/06/MATHEMATICS.pdf>
13. The New Kindergarten Curriculum in Ghana: Critiques and Analysis on Cultural Identity and Global Citizenship as a Core Competence - RSIS International, accessed June 3, 2026, <https://rsisinternational.org/journals/ijriss/Digital-Library/volume-5-issue-3/89-94.pdf>
14. NATIONAL PRE-TERTIARY EDUCATION CURRICULUM FRAMEWORK, accessed June 3, 2026, <https://nacca.gov.gh/wp-content/uploads/2019/04/National-Pre-tertiary-Education-Curriculum-Framework-final.pdf>
15. MINISTRY OF EDUCATION REPUBLIC OF GHANA KINDERGARTEN CURRICULUM (KG 1&2) SEPTEMBER 2019 Kindergarten Curriculum for Prescho, accessed June 3, 2026, <https://nacca.gov.gh/wp-content/uploads/2019/06/KG-Curriculum.pdf>
16. Teachers' Concerns on the Implementation of the Standards-Based Curriculum in Ghana: A Case of New Juaben North Municipal - SCIRP, accessed June 3, 2026, <https://www.scirp.org/journal/paperinformation?paperid=125414>
17. NATIONAL COUNCIL FOR CURRICULUM & ASSESSMENT, accessed June 3, 2026, <https://mingycomputersgh.wordpress.com/wp-content/uploads/2023/10/social-studies.pdf>
18. NATIONAL COUNCIL FOR CURRICULUM & ASSESSMENT, accessed June 3, 2026,

- https://nacca.gov.gh/wp-content/uploads/2024/04/Social-Studies_CCP_JHS-1-3_Revised.pdf
19. Ghana's Common Core Mathematics Curriculum | PDF | Educational Assessment - Scribd, accessed June 3, 2026, <https://www.scribd.com/document/982840478/basic-7-maths>
 20. TRAINING GUIDE SCIENCE, accessed June 3, 2026, <https://mingycomputersgh.wordpress.com/wp-content/uploads/2019/09/science-training-guide-june-2-2019.pdf>
 21. Top Payment Gateways in Ghana - Dusupay, accessed June 3, 2026, <https://www.dusupay.com/post/top-payment-gateways-in-ghana>
 22. Payment Channels | Paystack Developer Documentation, accessed June 3, 2026, <https://paystack.com/docs/payments/payment-channels/>
 23. Mobile Money Integration for Ghana Websites — Faciotech Blog, accessed June 3, 2026, <https://blog.faciotech.com/mobile-money-payment-integration-ghana>
 24. Overview – NaSIA | National Schools Inspectorate Authority, accessed June 3, 2026, <https://www.nasia.gov.gh/registration/overview/>
 25. All private schools are required to be licensed by 2024 - NaSIA - CitiNewsroom.com, accessed June 3, 2026, <https://www.citinewsroom.com/2023/10/all-private-schools-are-required-to-be-licensed-by-2024-nasia/>
 26. NaSIA extends school licensing and registration to December - Ghana Web, accessed June 3, 2026, <https://www.ghanaweb.com/GhanaHomePage/NewsArchive/Nasia-extends-school-licensing-and-registration-to-December-1239994>
 27. SCHOOL ESTABLISHMENT AND INSPECTION POLICY (SEaIP), accessed June 3, 2026, https://www.nasia.gov.gh/wp-content/uploads/Nasia_Final-SEaIP_DR.HHA.YAA.REVIEW_30APRIL2021_CLEAN-VERSION.pdf
 28. GUIDELINES FOR ESTABLISHING A NEW SCHOOL (PART A) NaSIA issues Approval Letter, accessed June 3, 2026, https://www.nasia.gov.gh/wp-content/uploads/GUIDELINES-FOR-ESTABLISHING-A-NEW-WEbVISION_SCHOOL_20APR2021_V1.0.pdf
 29. Data Privacy Policy - Metaschool AI - Accredited AI-Powered Online ..., accessed June 3, 2026, <https://www.metaschoolai.com/privacy/>
 30. Data protection laws in Ghana, accessed June 3, 2026, <https://www.dlapiperdataprotection.com/index.html?t=law&c=GH>
 31. Digital Regulations in the Metaverse Era - GHANA - International Bar Association | IBA, accessed June 3, 2026, <https://www.ibanet.org/document?id=Digital-Regulations-in-the-Metaverse-Era-Ghana>
 32. Data Protection Commission: Home, accessed June 3, 2026, <https://dataprotection.org.gh/>